



Guideline on AT32 MCU hardware design and ESD/EFT immunity design(未完)



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1. Power supply

For ensuring full-speed running and full-featured implementation, AT32 microcontroller requires an operating voltage supply (VDD) which is defined in the appropriate data sheet. An internal LDO is used to supply the 1.2V digital power. The Real-time clock (RTC) and battery powered registers (BPR) can be powered by the VBAT when the VDD is powered off.

1.1 Independent ADC/DAC/CMP supply and reference voltage

To improve ADC/DAC conversion accuracy and comparator (CMP) robustness, an independent analog power supply pin (depending on the packages) is provided. With this pin, the ADC/DAC/CMP supply and reference voltage can be separately filtered and shielded from noise.

- ADC/DAC/CMP voltage supply is sourced from VDDA pin

- A separate supply ground pin is provided on the VSSA
- To allow for a better accuracy at low voltage inputs and outputs, the VREF+ (if available on packages) can be connected to a separate external reference voltage that is below the VDDA. However, such reference voltage must be greater than the minimum VREF+ value.
- VREF+ can be connected to VDDA. The VREF- must be connected to VSSA. They are internally connected to VDDA and VSSA respectively when the VREF+ and VREF- are not available.

1.2 Battery power supply

To retain the content of the BPR registers and ensure RTC running when the VDD is turned off, the VBAT pin can be connected to a battery or another backup power supply. If no external battery is used in the application, then the VBAT must be connected to VDD externally.

1.3 Power supply scheme

- VDD pin must be connected with an external decoupling capacitor (it can be a separate tantalum or ceramic capacitor of at least 4.7 μF . And each single VDD must be connected with a 0.1 μF ceramic capacitor)
- VBAT pin can be connected to an external battery. If no external battery is used, VBAT pin must be connected to VDD with a 0.1 μF ceramic capacitor
- VDDA must be connected to two external decoupling capacitors (1 μF tantalum or ceramic capacitor + 0.1 μF ceramic capacitor)
- VREF+ pin can be connected to VDDA. If VREF+ is to be linked to an external reference voltage separately, then it must be connected with two decoupling capacitors (1 μF tantalum or ceramic capacitor + 0.1 μF

ceramic capacitor). The range of VREF+ must comply with the requirements defined in the “ADC/DAC characteristics” of the datasheet.

- More measures can be used to shield the analog power supply from noise, such as, by adding magnetic beads on the front of VDDA/VREF+

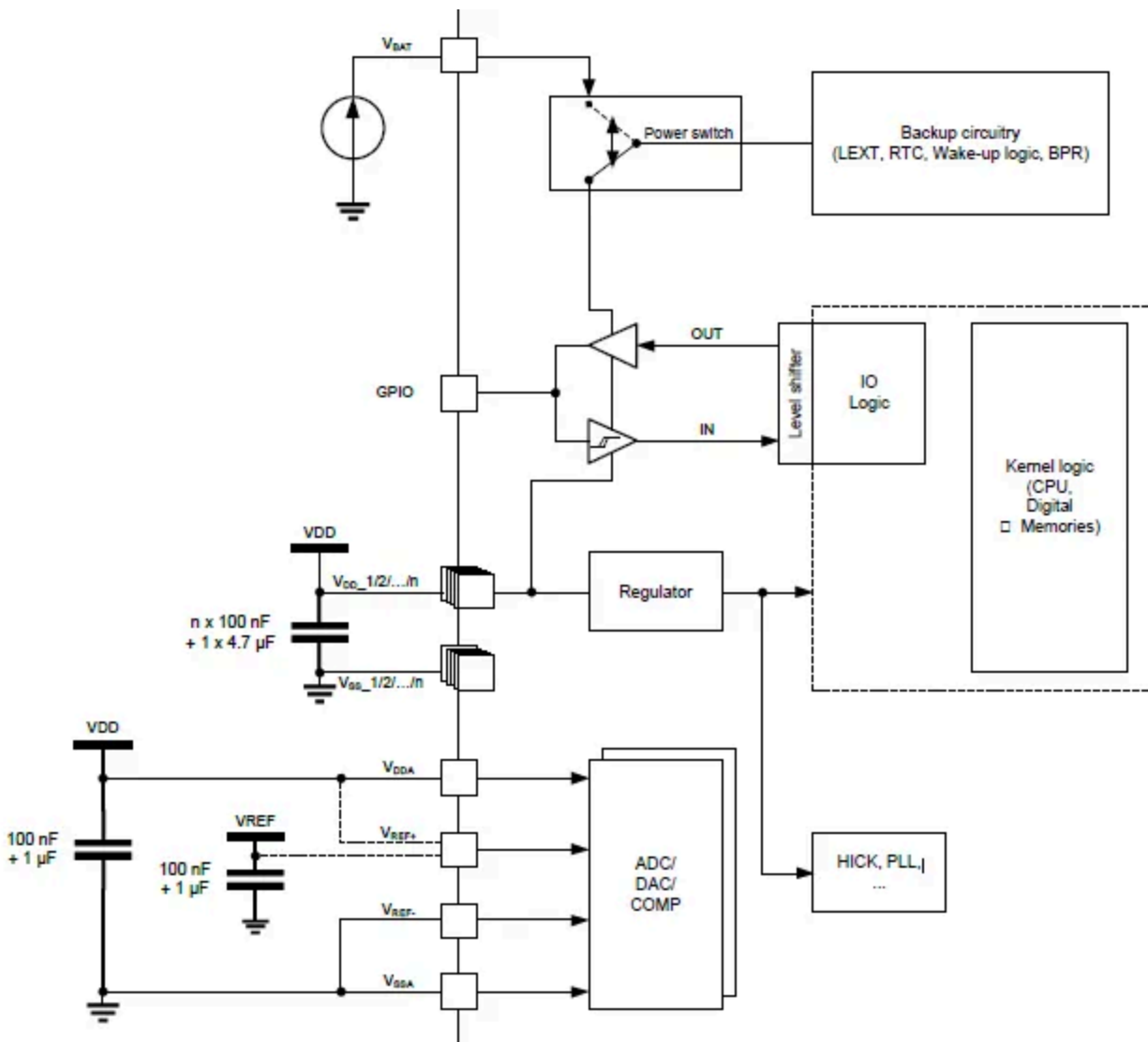


Figure 1. Power supply solution

1.4 Reset and power supply monitoring

1.4.1 Power-on reset (POR)/low-voltage reset (LVR)

The AT32 MCU has an integrated POR and LVR circuitry which allows its proper operation when the VDD voltage reaches VPOR. The device remains

in Reset state when VDD is below a specified threshold VLVR, without the intervention of external reset circuit. For more details about the POR/LVR reset threshold, refer to the electrical characteristics of the datasheet.

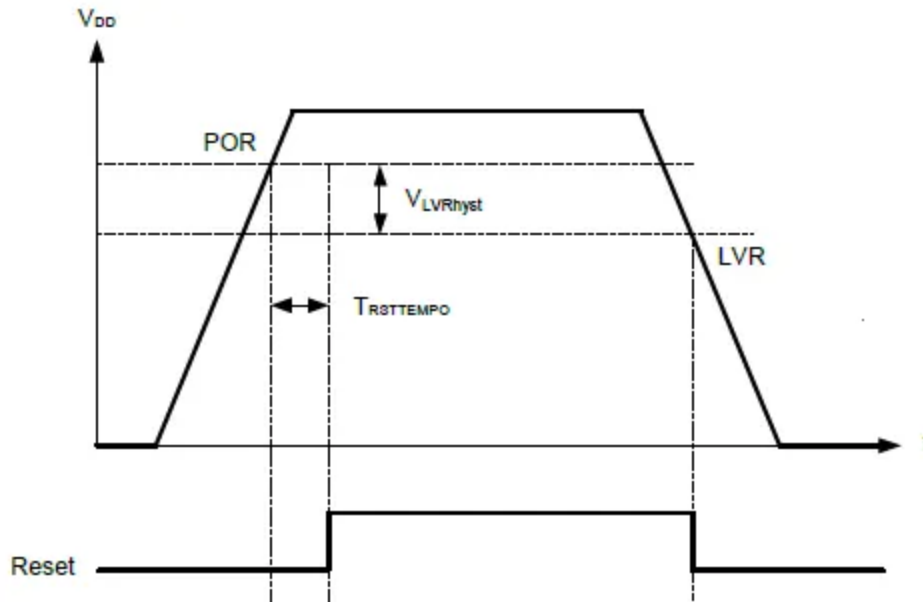


Figure 2. POR/LVR waveform

1.4.2 Power voltage monitoring (PVM)

The user can use the PVM to monitor the VDD voltage by comparing it to the threshold selected by the PVMSEL bit in the power control register (PWC_CTRL).

The PVM is enabled by setting the PVMEN bit. The PVMOF bit in the power control/status register (PWC_CTRLSTS) is used to indicate whether VDD is higher or lower than the selected PVM threshold. This event is internally connected to the external interrupt line 16 and can generate an interrupt if it is enabled through the external interrupt register. The PVM interrupt will be generated when VDD drops below the PVM threshold and/or when VDD rises above the PVM threshold, depending on the external interrupt line 16 rising/falling edge setting. A case in point is that this feature can be used to perform emergent shutdown tasks.

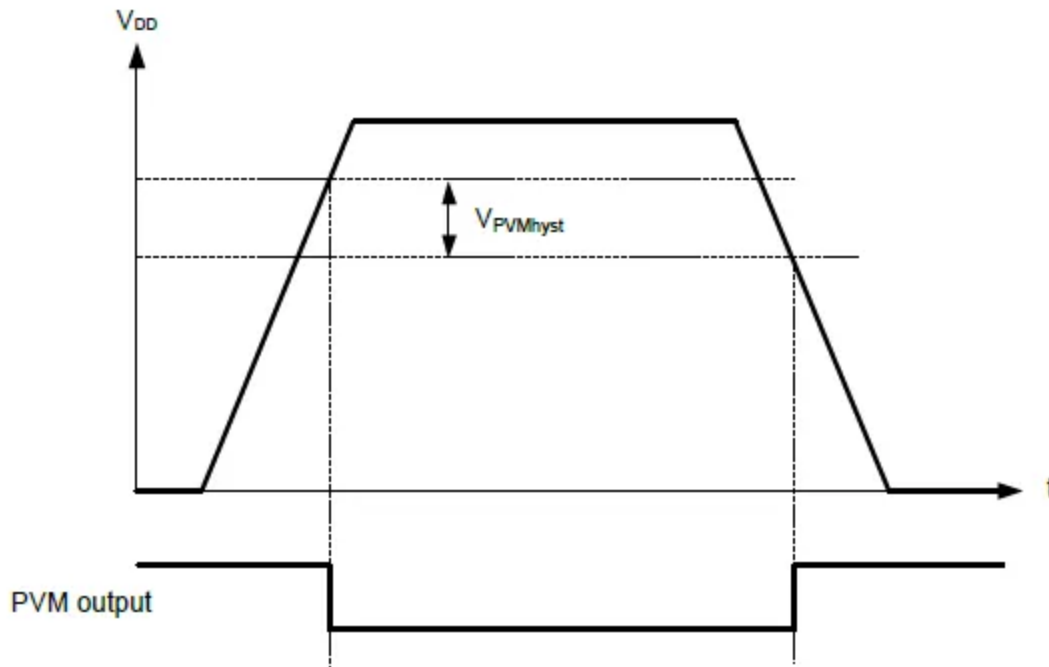


Figure 3. PVM thresholds and output

1.4.3 System reset

A system reset will set all registers to their reset values except the reset flags in the clock controller CRM_CTRLSTS register and battery-powered registers.

A system reset is generated when one of the following events occurs:

- VDD drops below VLVR
- A low level on NRST pin (external reset)
- Watchdog end of count (WDT reset)
- Window watchdog end of count (WWDT reset)
- Software reset

- Low-power management reset

The sources of reset events are indicated by reset flags in the CRM_CTRLSTS register.

1.4.4 NRST circuitry

It is highly recommend to connect the NRST pin to a 100 nF capacitor for better EMC performance. The NRST is internally pulled up to VDD. A typical pull-up resistance is 40 kΩ, thus removing the need of an external pull-up resistor.

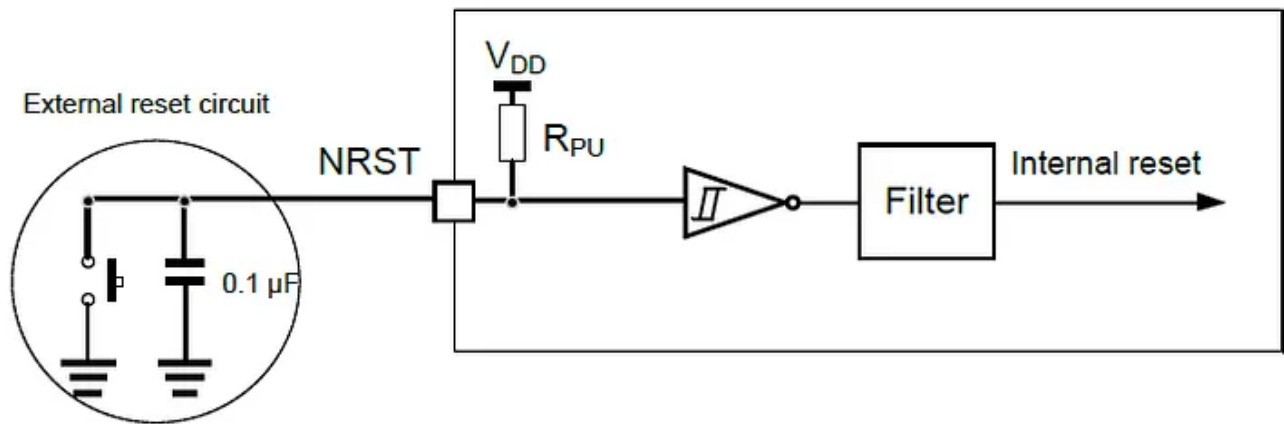


Figure 4. NRST circuit

2. Clocks

Three different clock sources can be used to drive the system clock (SCLK)

- HEXT high-speed external oscillator clock
- HICK high-speed internal oscillator clock
- PLL clock; PLL can be clocked by the HEXT or HICK

The devices have the following two secondary clock sources:

- Low-speed internal clock (LICK) which drives the watchdog and the RTC clock (RTCCLK) optionally used for waking up the system from Deepsleep/Standby mode
- Low-speed external crystal (LEXT) which optionally drives the RTC clock. Any clock source can be turned on or off independently when it is not used, in order to optimize system power consumption.

2.1 HEXT clock

The high-speed external clock signal (HEXT) can be generated from external clock source or crystal/ceramic resonator.

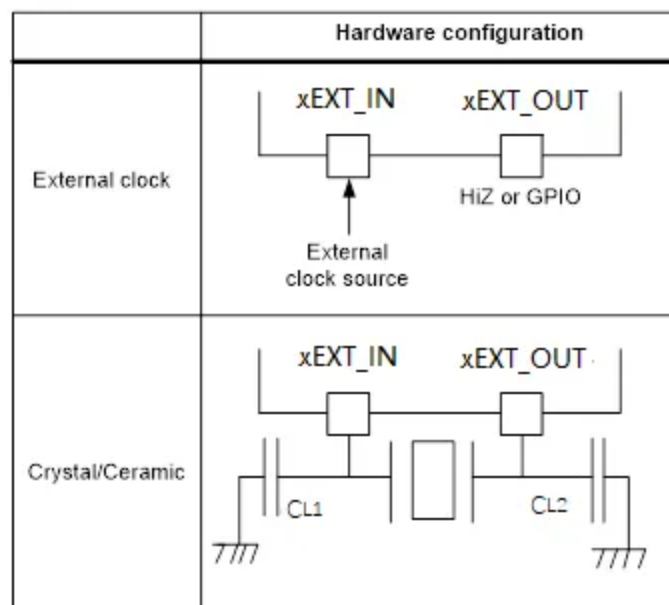


Figure 5. HEXT/LEXT clock sources

If external clock source mode is selected, an external clock must be provided and its frequency can be up to 25 MHz. The external clock signal (square, sinus or triangle with 45 ~ 55 % duty cycle) must be connected to the HEXT_IN pin while the HEXT_OUT pin should be left floating (some new

products support the use of HEXT_OUT as GPIO simultaneously), see Figure 5.

When crystal/ceramic resonator is used as a clock source:

4 ~ 25 MHz external crystal can provide more accurate main clock for the device. The crystal/ceramic resonator and the loading capacitors have to be placed as close as possible to the oscillator pins in order to minimize output distortion and startup stabilization time. The loading capacitance values must be adjusted according to the selected oscillator. See Figure5.

The crystal manufacturer typically specifies a loading capacitor CL, which is the series combination of CL1 and CL2. $CL = CL1 \times CL2 / (CL1 + CL2) + C_{stray}$, in which the Cstray is a PCB and pin capacitance, in the 2 pF ~ 7 pF range.

For 8 MHz / 20 pF oscillator, its drive level (DL) has a typical value of around 100 uW. If using a higher-frequency oscillator, its drive level will also be greater. When it comes to the reduction of the drive level, AT32L series is different from AT32F series in terms of operating methods. For AT32L series devices, you can select a lower drive level through related registers; For AT32F series devices, the only option is to connect them to external resistors in parallel or in series connection so as to reduce the drive level. The true drive level of an oscillator depends on the measurement of oscillator vendors.

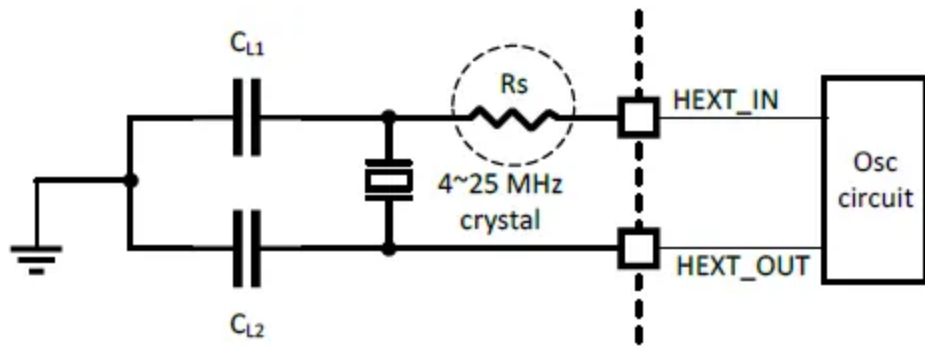


Figure 6. HEXT series resistor diagram

It should be noted that if MCU OTG master mode or CAN feature is used, for a better clock accuracy, the PLL which is generated by HEXT should be used to clock OTG master mode or CAN module. For USBFS or OTGFS device mode, they can be clocked by the HICK 48 MHz to save HEXT-related components and routes.

For more informations, refer to the electrical characteristics of the datasheet or reference manual.

2.2 LEXT clock

The low-speed external crystal signal (LEXT) can be generated external clock source or crystal/ceramic resonator.

The LEXT clock is a 32.768 kHz low-speed external crystal or ceramic resonator. It is capable of providing a low-power but highly accurate clock source to the real-time clock or other timing functions.

When crystal/ceramic resonator is used as clock source, the crystal/ceramic resonator and the loading capacitors have to be placed as close as possible to the oscillator pins in order to minimize output distortion and startup stabilization time. The loading capacitance values must be adjusted according to the selected oscillator. See Figure 5.

If external clock source mode is selected, its maximum frequency must not exceed 1 MHz. The external clock signal (square, sinus or triangle with 45 ~ 55 % duty cycle) must be connected to the LEXT_IN pin while the LEXT_OUT pin should be left floating (some new products support the use of LEXT_OUT as GPIO simultaneously), see Figure 5.

For more informations, refer to the electrical characteristics of the datasheet or reference manual.

3. Boot mode

Three different boot modes can be selected through the BOOT [1:0] pin (depending on product model, it may be an external BOOT1 pin or the reverse value of nBOOT1 in the user system data) as shown in Table 1:

Boot mode	Description	Configuration	
		BOOT1	BOOT0
Main Flash memory	Main Flash memory is selected as the boot space	X	0
Boot memory	Boot memory is selected as the boot space	0	1
SRAM	Embedded SRAM is selected as the boot space	1	1

Table 1. Boot modes

The values on the BOOT pins are latched after a system reset. The user can select the required boot mode by setting the BOOT0 and BOOT1 pins after a reset. The BOOT values are re-latched when the device leaves the Standby mode. Therefore, the BOOT pins must remain in the required boot mode configuration when the device is in Standby mode.

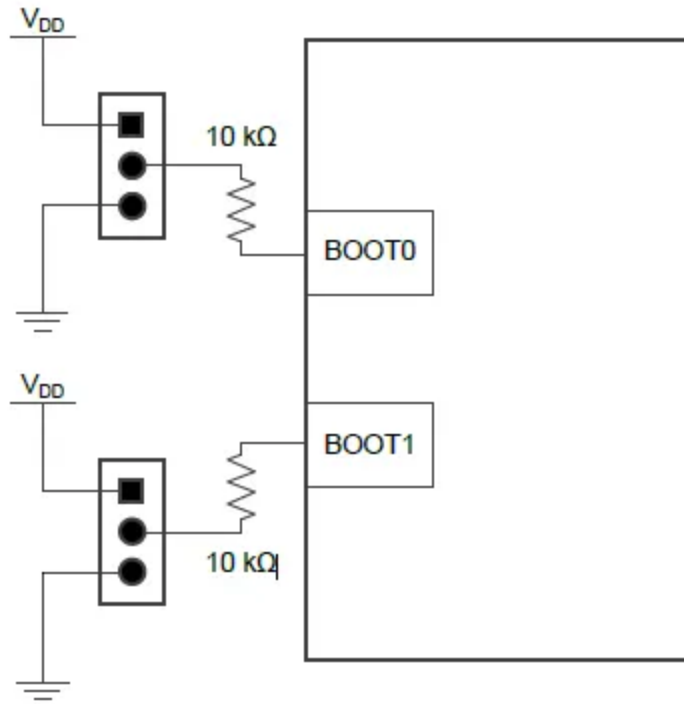


Figure 7. Boot mode selection

4. Debug port

The AT32 MCU integrates the Serial Wire/JTAG debug port.

- Serial Wire Debug port (SWD)
- Standard JTAG Debug port (JTAG). Some products do not support the full functionality of JTAG, but SWO feature is still available.

Debug port pin name	JTAG debug port		SWD + SWO debug port		Pin assignment
	Type	Description	Type	Description	
JTMS/SWDIO	I	JTAG test mode selection input	I/O	Serial Wire data input/output	PA13
JTCK/SWCLK	I	JTAG test clock input	I	Serial Wire clock input	PA14
JTDI	I	JTAG test data input	-	-	PA15
JTDO/SWO	O	JTAG test data output	O	Asynchronous trace	PB3
JNTRST	I	JTAG test reset	-	-	PB4

Table 2. Debug port pin assignment

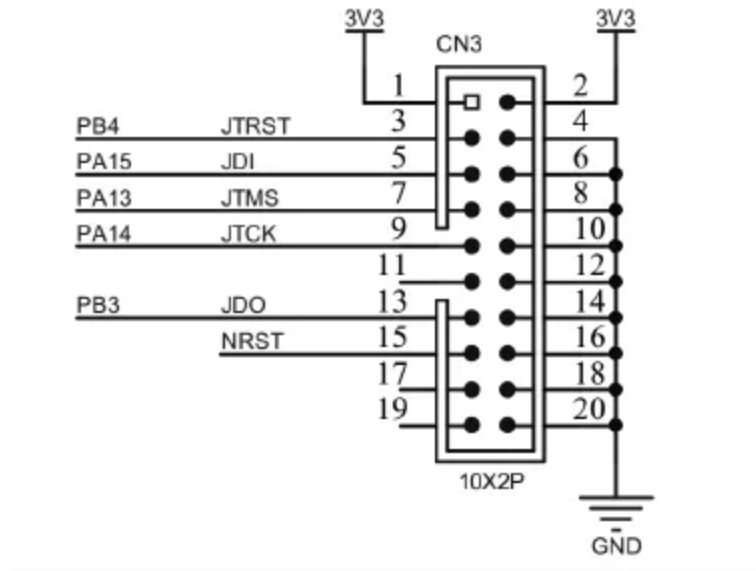


Figure 8. JTAG standard port circuit

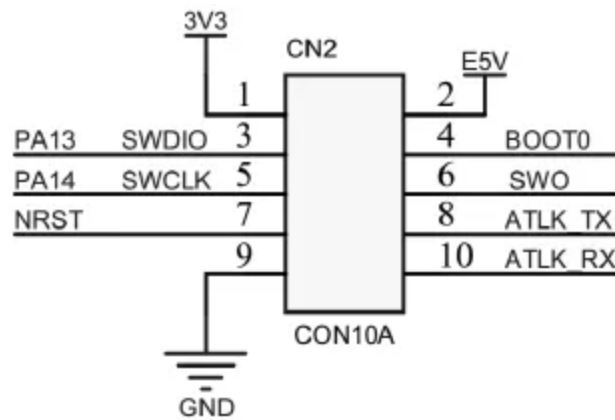


Figure 9. Combining SWD with AT-Link

5. Reference design

Figure 9 below takes AT32F421 series as an example. For more design information, refer to the AT-START evaluation board of the corresponding AT32 MCU series.

Items	Description	Y, N, N/A
Power supply	Is V _{DD} pin voltage within spec?	☐
	Are the voltages on V _{DDA} /V _{REF+} pins within spec?	☐
	Are power pins connected with decoupling capacitors?	☐
	Is the input voltage on V _{BAT} pin within spec?	☐
	Is the V _{BAT} pin connected to V _{DD} if no external voltage source is used ?	☐
Clocks	Is an external active clock signal input on HEXT_IN pin?	☐
	Are the external active/passive clock frequencies within spec?	☐
	Is the duty cycle of an external clock in the 45% to 55% range?	☐
	Does the loading capacitance of a resonator comply with the manufacturer's spec? Are MCU pin capacitances taken into account?	☐
	When OTG master mode or CAN feature is to be used, is there a HEXT crystal circuit as a system clock?	☐
Reset	Is the NRST pin connected with 0.1 μ F ceramic capacitor?	☐
Program and debug	Is the SWD/JTAG port connection correct?	☐
GPIO pins	Are PC13/PC14/PC15 used as output pins with less than 3 mA drive current?	☐
	Are all the sink/sourcing currents on GPIO pins less than 25 mA?	☐
	Are the total amount of sink/sourcing currents on GPIOs less than 150 mA?	☐
	Is the input signal level on GPIO pins within spec?	☐
	Are FT pins not used when they are supposed to be? (such as 5 V CAN transceiver)	☐
ADC	Is ADC input signal less than V _{REF+} ?	☐
	Are the input impedances of ADC sample channels within spec?	☐

Table 3. Schematic checklist

7. Hardware design tips

It is recommended that the users follow the rules described in the subsequent sections for circuit design and PCB layout in order to achieve better device performance and lower noise disturbance.

7.1 General PCB design tips

The layout design of PCB should take into account the placements of various functionality circuits and devices. For example, weak signal analog circuit, high-speed digital circuit and large noise circuit have to be separated from one another during PCB layout in order to minimize coupling interference from circuits. When it comes to the placements of devices, it is important to check if there is crossing or mixing issue among signal routes of different circuits.

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- Voltage regulator and power supplies should be placed near power input port. Do not route the power signal line and high noise signal line in parallel.
- Sensitive devices such as microcontroller and crystal should not be placed PCB edge; Instead, connection interfaces should be placed PCB edge.
- Components within a functional module should be placed as close as possible to each other, so as to minimize their physical distances in between.

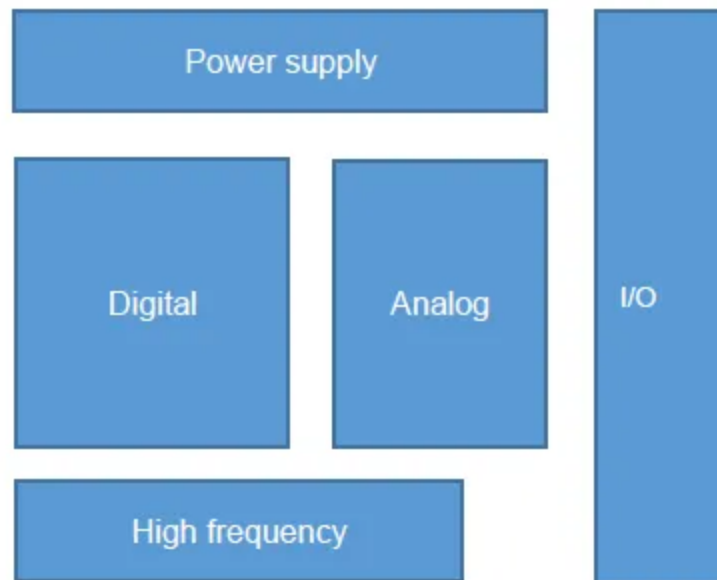


Figure 11. Placement of functional module

7.1.1 Power supply and ground

In most cases, the VDD/VSS pins should be routed in star-like mode (right below), not in chain-alike mode (left below). For multi-layer board, the VDD pin is directly connected to internal power supply, while VSS pin is directly connected to internal ground plane. It is recommended to place a digital ground plane just below the microcontroller.

Here are some recommendations for better EMC performance:

- Use multi-layer board
- Place one or two decoupling capacitors adjacent to power supply pins to reduce possible disturbance
- Use parallel capacitors set to expand the frequency range for filtering noise
- Ensure that only one star like node connects the analog ground with digital ground. This node should be placed as far as possible to MCU, but

as close as possible to the bypass capacitor of a linear power supply regulator

- Isolate analog input signals through analog ground
- Minimize power and ground loop areas
- Ensure that at least two vias are available to connect power lines on different layers
- Apply protective components to the corresponding pins or port lines

7.1.2 Decoupling capacitor

Decoupling capacitors must be as close to the VDD pins as possible for the purpose of decoupling.

A μF -level bypass capacitor should be placed at MCU's power input port (that is, along the line from PCB's LDO to MCU's power input port) to buffer voltage variations from public power lines and the noise from bypass power supplies. If a small ceramic capacitor and a large electrolytic capacitor are connected in parallel for decoupling purpose, the former (ceramic capacitor) should be placed as close to the VDD pin as possible to minimize loop areas.

If possible, it is advised to place all decoupling capacitors on the same layer as MCU, or put them at the bottom layer under pairs of power pins (VDD/VSS).

Power routing should first run through a decoupling capacitor pad and to VDD pin so as to filter noise currents.

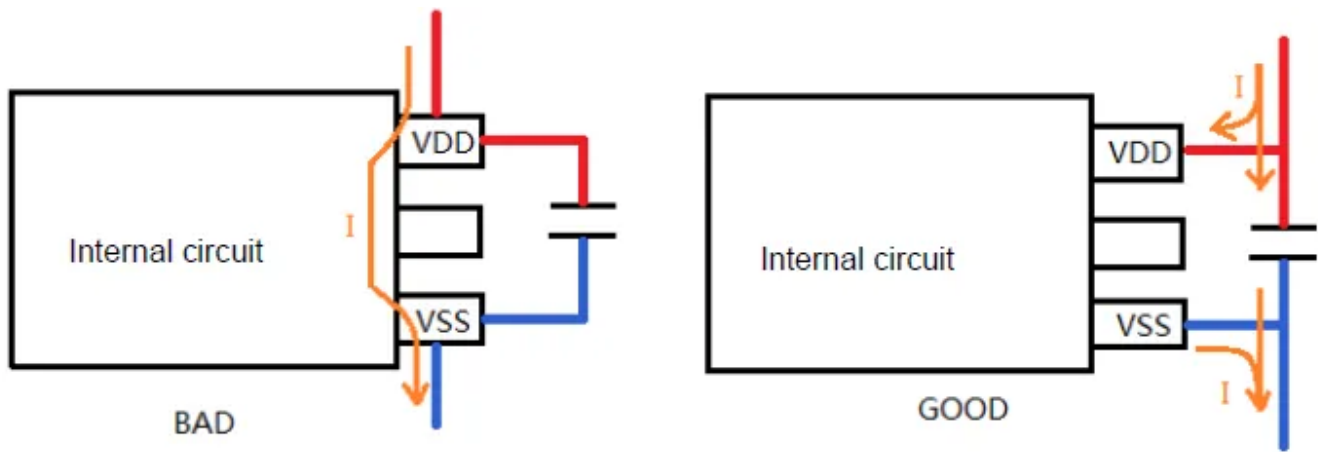


Figure 13. Decoupling capacitors layout

Power vias should not be placed between a decoupling capacitor and VDD pin as this otherwise would invalidate the decoupling capacitor.

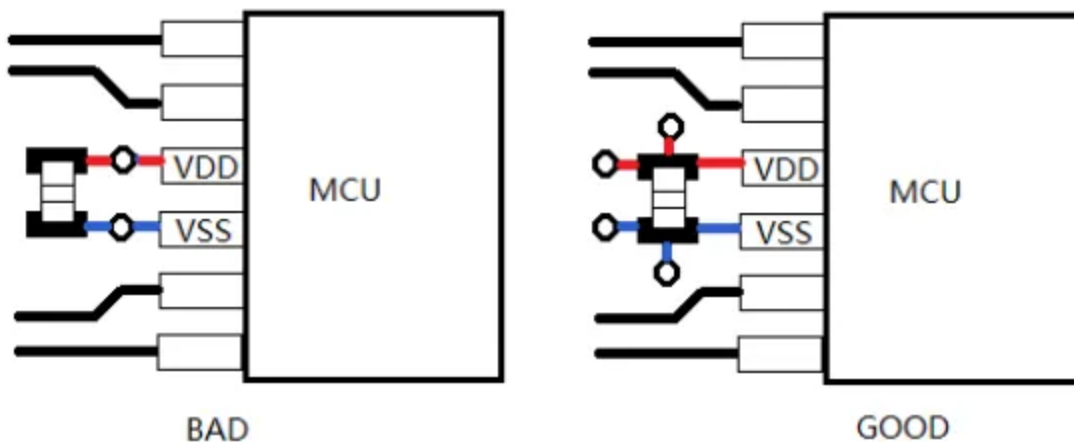


Figure 14. Vias location

7.1.3 Signal routing

After the completion of power and ground routing, priority should be given to crystal routing. Mixed or crossed routing should be avoided when there are mixed digital and analog signals in a circuitry.

The layout of sensitive signals should be handled in the first place, and they should be protected against interference. For MCU peripheral circuits, the sensitive signals include clocks, reset and interrupt signal lines.

- Minimize the length of parallel routing
- If crossed routing is required for signals, they should be overlapped vertically to minimize decoupling capacitances arising from signals
- The distances among signal lines should comply with “3W” rule to minimize decoupling inductances

Additionally, USART/UART receive mode is relatively sensitive to the design of signal sampling. In most cases, they are normal in operation. But the start bit is likely to be deemed as invalid in severe disturbance environment so that data reception error may occur. As a result, it is recommended that a capacitor of 1~10 nF be grounded at RX pin, as close as possible to MCU side, with the aim of avoiding interference.

7.1.4 Crystals

The crystal oscillator is highly sensitive to disturbances at startup time. It is highly recommended that the crystal signal lines should not be placed in parallel with high current switching lines.

The general rules are as follows:

- A loading capacitor should be as close to the HEXT_IN/HEXT_OUT as possible and is protected with ground rings
- Signal lines (except ground line) should never be routed near crystal circuit or across the bottom-layer circuit
- Select a crystal with minimum frequency that meet system requirements
- Use a suitable loading capacitor
- Ground the crystal oscillator shell

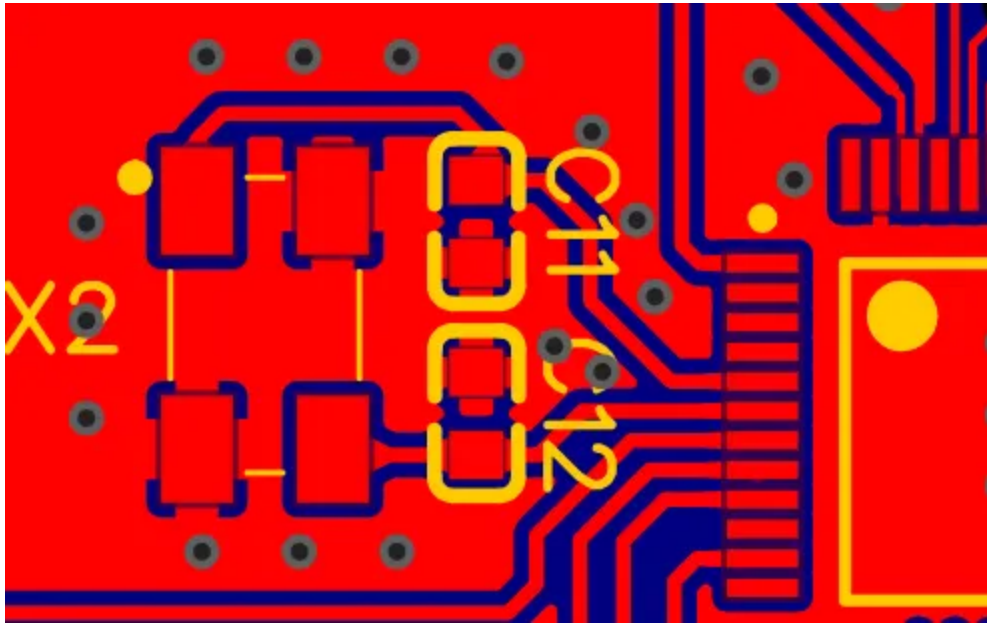


Figure 15. Crystal layout

7.1.5 Analog signals

A weak analog signal is highly susceptible to digital signal. If both signals are required to be mixed, it is important to ensure that they are routed in an orthogonal way to minimize coupling effect.

If the VDDA/VSSA and VREF+/VREF- are not isolated from digital power and digital ground or not filtered, the ADC performance will be severely impacted.

7.1.6 Input/output ports

GPIOs in input mode are more sensitive to noise than in output mode.

To reduce external noise, it is advised to add a RC filter onto GPIO port that is configured as input mode. The filter should be placed as close to the GPIO pins as possible.

RC value depends on the characteristics of input signals. Typically, series resistance value is in the range of $100\ \Omega \sim 1\ \text{k}\Omega$, and filtering capacitance 1

nF $\sim$ 0.1 μ F.

The layout where the NRST pin is externally connected to decoupling capacitors and pull-up resistors should be regarded as filter for power pins. As a result, the groundings loops of capacitors and VDD loops of pull-up resistors should be minimized.

For unused GPIO pins:

- If GPIO pins remain floating, configure them as output low by software
- If GPIO pins are not floating, connect unused pins to a 10 k Ω pull-down resistor
- It is not recommended to connect unused GPIO pins directly to power supply or ground

7.2 QFN packages EPAD design tips

The QFN (that is Quad Flat No-lead) package is a surface mount chip scale package with small pad size and low profile. The QFN package, characterized by lightness, thinness, shortness and low profile, providing lower and competitive price. It also has the advantage of saving PCB area. In particular, its lead-less design allows itself to be free from concerns arising from pin damage caused during packaging, transportation and production, which often occur in lead package.

QFN main features:

- Surface mount package
- Light weight, suitable for portable applications

- Use smaller PCB area
- Good thermal performance with soldering pad at the bottom of the package

7.2.1 Thermal via design

There is a large area of exposed pad at the bottom center of QFN package. This pad is not only used to connect chip to ground, but also for dissipating thermal generated during chip running. For the same product, the θ_{JA} value with QFN package is usually lower than that of QFP or SOP package. In most cases, the EPAD is directly soldered onto a circuit board. The vias in EPAD help dissipate excessive thermal into copper ground layer.

For multi-layer PCB, the thermal can be transmitted to other copper layer through vias, thereby enhancing the capacity of chip to dissipate thermal.

7.2.2 Thermal via dimensions and placement

Theoretically, the greater the number of vias, or the bigger via diameter, the better the thermal conduction it generates. In other words, using more vias with bigger diameter often produce better thermal conduction. However, too large diameter is very likely to cause solder leakage and thus affect solder reliability.

Recommendations are as follows for via dimensions:

- Via Pitch: 0.8~1.2 mm
- Via Diameter: 0.3 mm

Based on this, taking QFN32 4 x 4 mm package as an example, Figure 15 shows the dimensions and placement of vias on QFN packages.

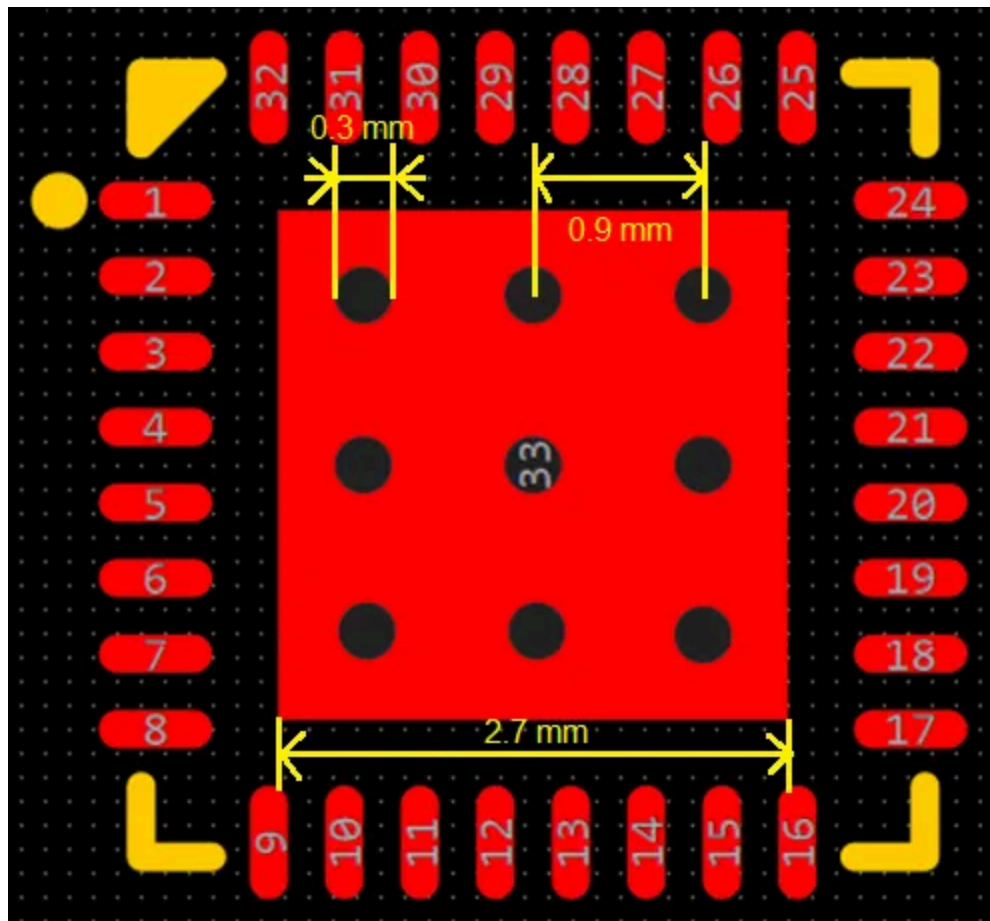


Figure 16. EPAD vias on QFN package

Taking AT32F435 QFN48 6 x 6 mm as an example, Figure 16 compares the temperature change when using different number of vias on EPAD. Under the same conditions, the internal chip temperature with 4 vias is higher than that with 16 and 25 vias.

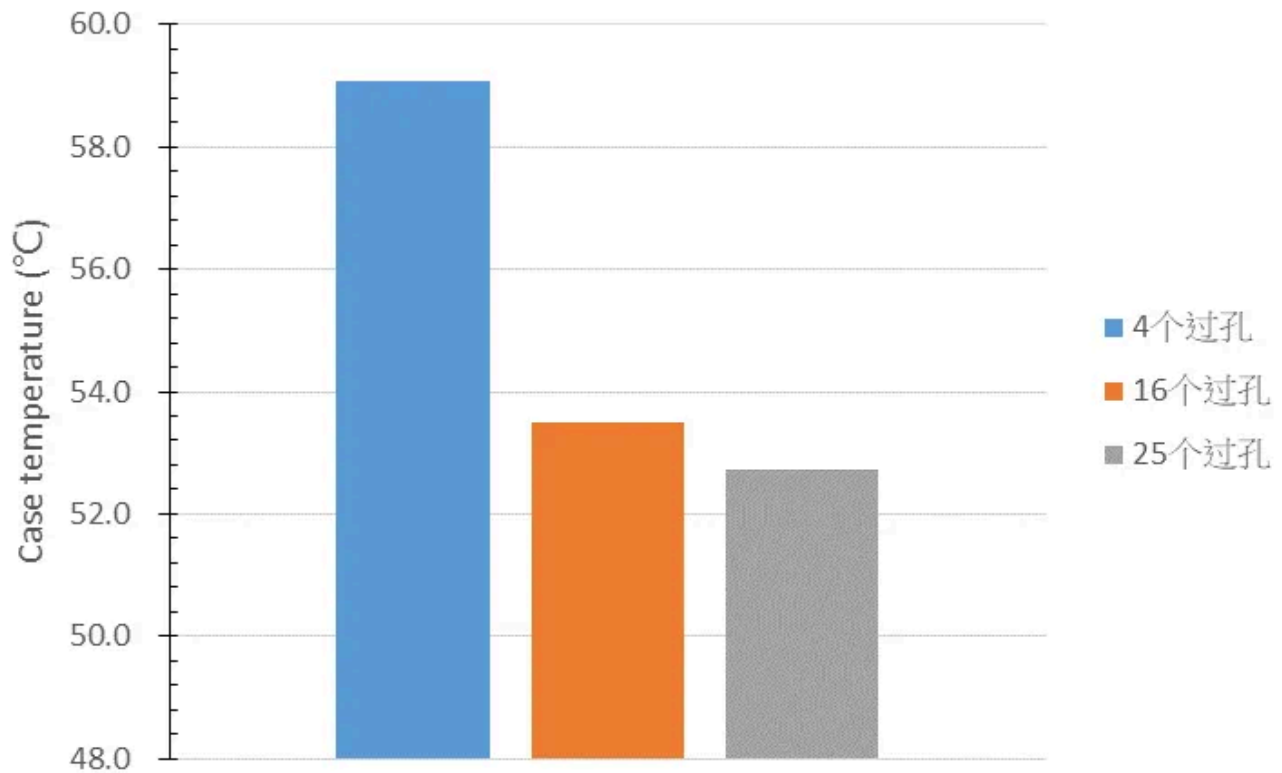


Figure 17. Temperature comparison with different number of vias on EPAD

7.3 Other design tips

A robust system design should take into account both hardware layout and software design. Good system design, in conjunction with hardware configuration, can allow for a quick correction of system failures when they occur, and provide strong system protection by means of preventive features. Thus, it is important to guarantee system reliability by ensuring a close cooperation between software and hardware. This section gives some proposals for user reference. For more information, refer to the document AN0041 or other related documents.

- Use watchdog to prevent system failure and code error:
When a microcontroller is disturbed, program pointer jumps to an unexpected location or enters infinite loop. In this case, the watchdog will restart the system allowing it to return to normal. It is advised to

refresh watchdog in main program rather than subprogram or interrupts.

- Define all interrupt vector functions, and add system reset sentences to the unused interrupt functions:
If unused interrupts are triggered by false signals, software will jump to the defined interrupt function, and resume microcontroller after performing a system reset
- Multiple-times access edge trigger event signals:
Read the register value of the trigger signal several times in the interrupt function to judge whether it is a jamming trigger signal. It is possible to add a random delay between two read operations for the evenly distributed jamming signals.

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